MILITARY SPECIFICATION

BAROSWITCH SIMULATOR ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of baroswitch simulator, entitled Baroswitch Simulator Assembly, which is used for training on the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8846202	- Baroswitch Simulator Assembly.
8846981	- Packing, Ammunition, for Simulator.
ET-8846202	- Equipment Tabulation.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The baroswitch simulator shall comply with all requirements specified on the drawing (dwg) 8846202, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 High potential leakage. The potential leakage shall not exceed 100 microamperes when tested as specified in 4.4.1.

3.6 Setting mechanism. The setting range of the baroswitch simulator assembly shall be from minus (-) 3000 feet to plus (+) 25,000 feet when tested as specified in 4.4.2. (One shaft rotation of the "fine" synchro transmitter is equivalent to 3600 feet of altitude. One shaft rotation of the "coarse" synchro transmitter is equivalent to 72,000 feet of altitude.).

3.6.1 Synchro calibration. The deviation of the fine synchro transmitter from electrical zero shall not exceed $\pm$ 1.5 degrees ($\pm$ 15 feet) when tested as specified in 4.4.2.4.

3.7 Resistance. The resistance of the two synchro rotors in parallel shall be 13 $\pm$ 2.6 ohms, and that of each circuit shall be within the tolerances specified when tested as specified in 4.4.3.1.

3.8 Expected life. Each synchro transmitter shall be capable of cycling from -3000 feet to 25,000 feet to -3000 feet for 3,000 cycles when tested as specified in 4.4.4.

3.9 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460(MU).

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

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3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.10.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8846202) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. High potential leakage (see 3.5)	X	
3. Setting mechanism (see 3.6)		X
4. Resistance (see 3.7)		X
5. Expected life (see 3.8)		X

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Stand-off (see dwg 9209538 covering a detail of dwg 8846202).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Overall length of stand-off (max.)	Gage
102.	Length of large diameter cylinder (max.)	Gage
103.	Diameter of end cylinders (max.)	Gage
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

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4.3.2.1.2 Upper and bottom plates (see dwgs 9209539 and 9209540 covering details of dwg 8846202).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of mounting holes (4 holes)	Gage
102.	Pitch diameter of threads	Gage
103.	Location of gear positional holes (8 for upper plate and 5 for bottom plate)	Gage
104.	Length of plates (max.)	Gage
105.	Width of plates (max.)	Gage
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.3 Baroswitch simulator assembly (see dwg 8846202).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Height of motor control above bottom plate (max.)	Gage
102.	Location of "coarse" transmitter gear below upper plate (min.)	Gage
103.	Any wire from motor and transmitters loose, broken, or missing	Visual/Manual
104.	Insulation sleeving loose, torn or damaged	Visual/Manual
105.	Solder loose, inadequate, or cold solder joint	Visual/Manual
106.	Any pin or plug connector bent	Visual
107.	Any pin missing	Visual
108.	Foreign matter on any pin, unless removable	Visual
109.	Connector deformed	Visual
110.	Resistor missing	Visual
111.	Lacing loose or missing	Visual/Manual
112.	Any gear improperly meshed with its mating gear	Visual
113.	Any gear improperly secured to its respective shaft	Visual/Manual

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Major:
114. Chips or other foreign matter in gear teeth.....Visual
115. Evidence of nicks and burrs in gear teeth.....Visual
116. Any stand-off loose or bent.....Visual/Manual
117. Objective evidence relative to material missing or improper (see 3.1).....Visual

Minor:
201. Markings missing, incorrect, or unidentifiable.....Visual
202. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.4 Carton, packing, ammunition, for baroswitch simulator, prior to sealing (see dwg 8846981).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:		
201.	Baroswitch simulator assembly loose	Manual
202.	Cushioning material missing or loose	Visual/Manual

4.3.2.1.5 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing, and packaging not in accordance with (see dwg 8846981)	Visual
102.	Marking not in accordance with (see dwg 8846981)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

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4.3.2.1.6 Carton, packing, ammunition, for baroswitch simulator, after sealing (see dwg 8846981).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major: None defined.		
Minor:		
	201. Fiberboard punctured, slit, or torn	Visual
	202. Marking missing, improper or illegible	Visual
	203. Label loose or missing	Visual/Manual
	204. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. The following test shall be performed:

Test	Test Classification
a. Setting mechanism	Major

This test is not considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Setting mechanism. Each baroswitch simulator shall be tested in accordance with 4.4.2. If any item fails to comply with 3.6, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). Els for assemblies are listed on the Equipment Tabulation, ET-8846202. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 High potential. The potential of the synchro and motor circuits shall be measured by utilizing a power supply of 500 ± 25 volts root mean square (vrms), 60 ± 5 Hertz (Hz) alternating current, and a megohm bridge, or equivalent. The potential shall be applied between pin T of the connector and unused pins B, C, D, L, M, P, N and V. The unused pins, tied together as a common terminal, may be connected to the connector guide pin or assembly support plate. Observation shall be made to determine that the leakage does not exceed 100 microamperes. Protective resistance shall be provided so that a reading of 1000 microamperes indicates a dead short.

4.4.2 Setting mechanism. The setting mechanism shall be tested by means of a 26 ± 1 volt, 400 ± 10 Hz alternating current power supply and a test cable. The power supply and capacitor (see 6.4) shall be connected to the motor and synchro circuits by means of the test cable.

4.4.2.1 Range setting limits. The setting mechanism shall be functioned to test for proper operation so that the remote reading at the ends of the setting range shall be -3000 feet and 25,000 feet.

4.4.2.2 Intermediate setting limits. Operate the setting system to the following intermediate settings: -1000 feet, 1000 feet, 4500 feet, 8000 feet, and 15,000 feet, within ± 10 feet. The intermediate settings shall be accomplished without the simulator exhibiting any drift, oscillation, or excessive roll. Drift is defined as the inability of the item to remain at the given setting after the unit has been stopped. Drift is caused by electrical power being continually applied to the motor. Oscillation is defined as having the unit travel alternately above and below the given settings. Excessive roll is defined as the inability of the item to come to rest in less than 100 ft. after release of the remote setting switch. Excessive roll is caused by excessive momentum in the gear train and lack of braking on the motor.

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4.4.2.3 Remote setting rate. The assembly shall be remotely set to a nominal altitude setting of 25,000 feet. The setting shall then be remotely changed to -3000 feet. The average setting rate shall exceed 5000 feet per minute.

4.4.2.4 Synchro calibration. Using the coarse synchro transmitter as a reference, the deviation of the fine synchro transmitter from electrical zero shall be determined by performing a test for calibration between the coarse and fine synchro transmitter at electrical zero and the point representing 00.000 altitude setting. The test device shall be calibrated against a zero-set transformer or synchro standard to within plus or minus 15 minutes.

4.4.3 Resistance.

4.4.3.1 Synchro rotor windings. The direct current resistance of two synchro rotors in parallel shall be measured by a method providing an accuracy of ± 5 percent, when compared with measurements made on a standard bridge. The rotor resistance for this test shall be measured at pins T and J of the connector. The resistance of each circuit measured at the pins of the connector shall be as follows:

Between pin	T	and	U	400 ohms	± 5%
"	"	T	A	400	"
"	"	T	H	400	"
"	"	T	F	400	"
"	"	T	K	10	"
"	"	T	R	10	"
"	"	T	E	10	"
"	"	T	S	10	"
"	"	A	H	40	"
"	"	K	R	10	"
"	"	E	S	10	"
"	"	E	J	25	"
"	"	R	J	25	"

The current during this test shall be limited to 300 milliamperes maximum and shall be applied for not more than 5 seconds.

4.4.4 Expected life. Apply 26 ± 2 volts, 400 ± 10 Hz ac power to the motor and synchro circuits of the baroswitch simulator assembly with a capacitor connected as specified in 6.4 and cycle between -3000 to 25,000 to -3000 feet for 3000 cycles. During cycling the motor current shall be interrupted at the ends of the setting range and a time period not exceeding 1 minute may elapse before reversing the direction of the drive.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The baroswitch simulator shall be packaged in accordance with dwg 8846981 using the specified moisture barrier.

5.1.2 Level C. The baroswitch simulator shall be packaged in accordance with dwg 8846981 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The baroswitch simulator shall be packed in accordance with dwg 8846981.

5.3 Marking. Marking shall be as specified in accordance with dwg 8846981.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Setting motor capacitor. Operation of the baroswitch simulator setting motor requires that a capacitor shall be connected across pins F and U of the electrical connector external to the baroswitch simulator. The capacitor used shall have a rating of 2.6 ± 5 percent microfarads at a working 200 volts dc.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2131 (Rev. B) dated 19 January 1966.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A199

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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